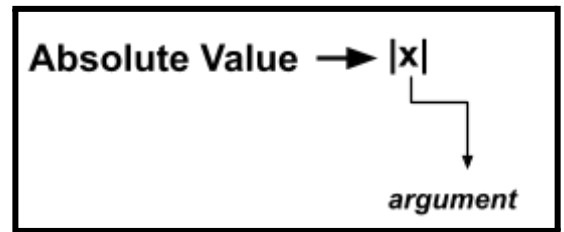


# Solving Absolute Value Inequalities

## Outcomes:

- Understand what absolute value means and how it affects inequalities.
- Solve absolute value inequalities by rewriting them without the absolute value bars.
- Represent solution sets using inequalities and intervals.



When solving inequalities with absolute values, you must remember:

$|x| < a$  means:  $-a < x < a$

$|x| > a$  means:  $x < -a$  or  $x > a$

## Hint:

“Less than” → inside (a range between two numbers).

“Greater than” → outside (two regions on the number line).

## Steps to Solve:

1. Isolate the absolute value expression

$|argument|$  should be alone on one side before moving on to the next step

2. Replace the absolute value sign with brackets and create two equations. The first stays positive and the second is negative.

$$|x| > a \rightarrow \begin{cases} (x) > a \\ -(x) > a \end{cases}$$

3. Solve using the rules of inequalities

| <b>Examples:</b> |                      |
|------------------|----------------------|
| $ x  < 5$        | $ x - 3  \geq 4$     |
|                  |                      |
| $2 x + 1  < 6$   | $-3 2x - 5  \leq -9$ |
|                  |                      |

### **Special Cases**

$|x| < -a \rightarrow$  no solution (absolute value cannot be negative)

$|x| > -a \rightarrow$  solution is all real numbers (absolute is always  $\geq 0$ )

| <b>Examples:</b> |                  |
|------------------|------------------|
| $2 x + 1  < -6$  | $-3 2x - 5  < 9$ |
|                  |                  |

## Zero Cases

$|x| < 0 \rightarrow$  no solution (absolute is always  $\geq 0$ ).

$|x| \leq 0 \rightarrow$  one solution (set the argument equal to 0 and solve).

$|x| > 0 \rightarrow$  solution is all real numbers except the value that makes it equal to 0.

$|x| \geq 0 \rightarrow$  solution is all real numbers (absolute is always  $\geq 0$ ).

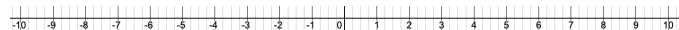
| <b>Examples:</b>          |                        |
|---------------------------|------------------------|
| $ x - 4  < 0$             | $ x + 2  - 3 \leq -3$  |
|                           |                        |
| $\frac{ x-4 }{2} + 2 > 2$ | $ 2x + 5  - 1 \geq -1$ |
|                           |                        |

## Remember:

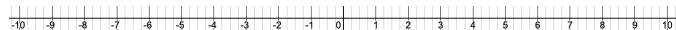
- Always **isolate the absolute value bars** first before splitting into cases.
- Remember the “less than = inside, greater than = outside” rule.
- Watch out for **negative numbers** on the right side. They can change the problem to “no solution” or “all real numbers.”
- Graphing the solution on a number line will help you visualize the intervals.

**You Try:**

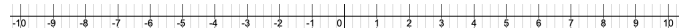
$$-3|x + 1| \leq -3$$



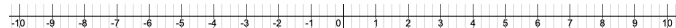
$$|x - 3| \geq 4$$



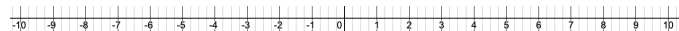
$$2|x + 1| < 6$$



$$-3|2x - 5| \leq -9$$



$$\frac{|x-2|}{3} < -1$$



$$\frac{|2x+1|}{4} \geq -2$$

